

The Decision Twin: Metaverse Patient Digital Twins as Executable Clinical Reality

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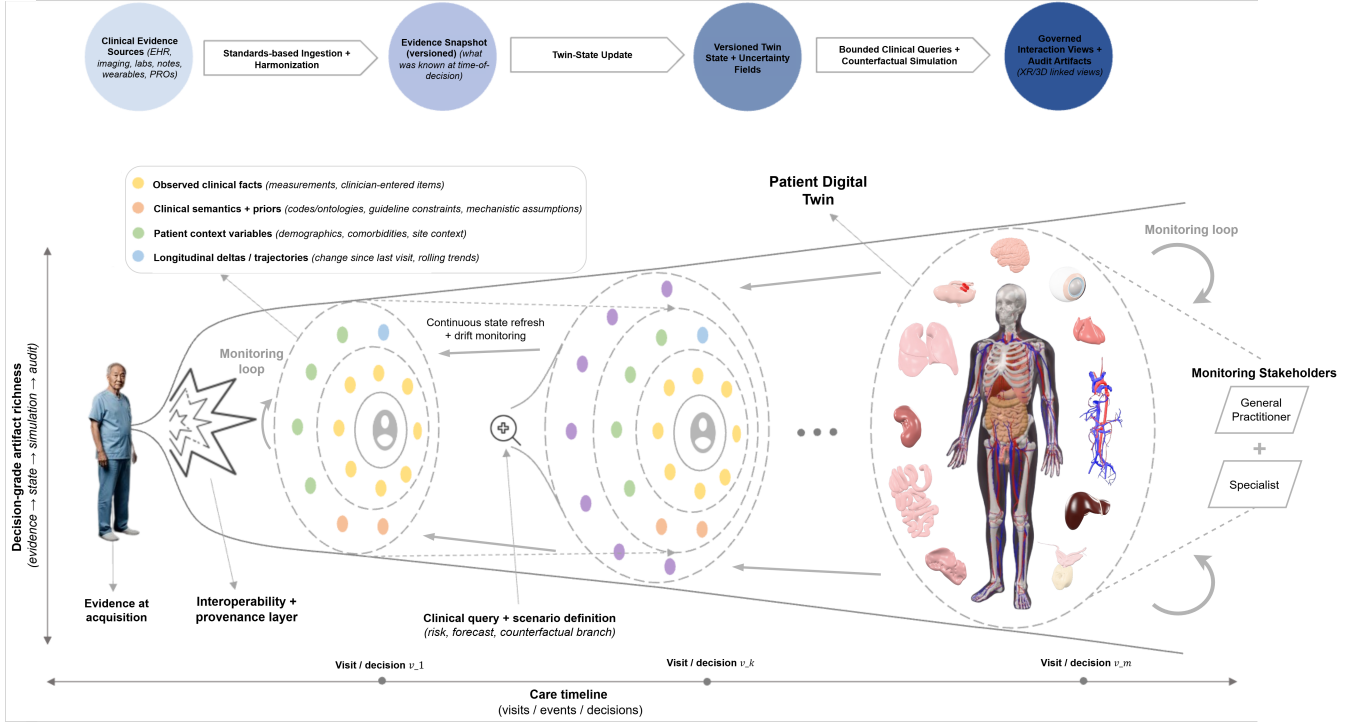


Figure 1: MPDT thesis: clinical AI becomes decision-grade when *state, simulation, explanation, and interaction* are bound into replayable, typed, auditable artifacts rather than delivered as disconnected products.

Abstract

Clinical AI is no longer bottlenecked only by model performance; it is bottlenecked by the *accountable interaction loop* through which clinicians and patients inspect evidence, test alternatives, and remain responsible for decisions under uncertainty. We argue that today’s digital-twin systems, XR interfaces, and foundation-model copilots each address part of this loop, but they fail when deployed as separate products: predictions are not replayable across time, XR becomes descriptive visualization without executable state, and copilots are fluent without auditable grounding. We introduce the *Metaverse Patient Digital Twin (MPDT)* as a decision-grade clinical artifact defined by one requirement: every displayed claim or simulated scenario must be traceable to a versioned patient state, explicit assumptions, and replayable interaction logs. We specify minimal acceptance criteria, a reference loop in which stakeholders observe new evidence, update the twin state, run bounded simulations, generate explanations, commit decisions, and then log and

monitor outcomes, along with a compact architecture that binds interoperability, simulation, governed interaction, and lifecycle controls. Finally, we outline workflows (risk stratification, diagnosis support, treatment rehearsal, training, cross-site coordination) and the evidence required to make MPDTs defensible: calibration over time, subgroup reliability, category-error prevention (observation vs. simulation), and measurable workflow outcomes.

Keywords

Patient digital twin; medical digital twin; extended reality (XR); metaverse healthcare; counterfactual simulation; explainable AI.

1 Introduction

1.1 The clinical paradox and why now

Modern healthcare is increasingly *data-abundant* but still *decision-constrained*: we can measure more, yet we cannot reliably *reconstruct and justify* what was known, assumed, simulated, and shown when a decision was made [31, 55]. Recent syntheses describe rapid

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growth in proposed applications—from precision health and disease management to therapy planning—while emphasizing that translation is limited by fragmented data, variable maturity, and uneven reproducibility [21, 42, 47, 51]. At the same time, immersive interfaces and foundation models expand how evidence can be communicated and interrogated, but they also amplify a core limitation: without a governed substrate, systems cannot reliably preserve *what was known, what was assumed, and what was simulated* at decision time [11, 23]. Figure 1 previews the central argument of the paper: the problem is no longer only how to model the patient, but how to make clinical reasoning inspectable, revisitable, and accountable once evidence, simulation, and interaction all enter the same decision process.

1.2 Why current paradigms fall short

Today, dominant paradigms each solve only part of the problem. *Digital twins* unify mechanistic knowledge and data-driven inference, but deployments often collapse into bespoke pipelines that are hard to validate, transfer, and audit [21, 31, 42]. *XR and metaverse applications* improve engagement and communication, but without an executable patient model they risk becoming descriptive dashboards rather than decision instruments [19, 40, 45]. *Foundation-model copilots* reduce documentation and synthesis burdens, yet remain unsuitable for ungoverned clinical decision-making due to instruction sensitivity, hallucination, and uneven reliability across tasks [1, 11, 50]. Beyond reliability, medically unjustified recommendation shifts across sociodemographic variations underscore why safety must be evaluated at the level of *workflow outcomes*, not benchmark accuracy [23, 36]. The deeper gap is infrastructural: clinical decision support cannot be safety-critical if evidence, state updates, simulations, and explanations cannot be traced, standardized, replayed, and monitored across time and sites. Interoperability standards (FHIR, DICOM, SNOMED CT, OMOP) define the minimum substrate for exchange and semantics, while FAIR principles become essential once derived artifacts enter longitudinal records [5, 12, 35, 49, 57]. Governance expectations for clinical AI further imply that credible systems must support controlled change, risk management, and computational-model credibility [7, 32, 52, 54]. Finally, human–automation research shows that confident, fluent, or visually compelling outputs can induce misuse/disuse even when evidence is weak [6, 24, 38]. The missing contribution is therefore not another model or interface, but a *replayable decision substrate*: a systems primitive that turns care into a versioned artifact graph (evidence → state → scenarios → interaction → outcome), enforcing epistemic separation between observed facts and simulated projections while remaining auditable under deployment.

1.3 Thesis, concept, and contributions

Our thesis is that personalized care is increasingly limited by *accountable interaction*, not representation alone. The critical failure mode is not only being wrong, but being wrong *without traceability*: when a system cannot reconstruct what evidence was ingested, how the twin state was updated, which counterfactual assumptions were activated, and which explanations were exposed to which stakeholder, it cannot be safely monitored, updated, or held accountable.

We propose the *Metaverse Patient Digital Twin (MPDT)* as a decision-grade patient twin whose defining property is an *auditable interaction loop*. Figure 1 makes this loop concrete as a multi-stakeholder system (patient, GP, specialist team, and the MPDT services): evidence is harmonized into a snapshot, committed into a versioned state, queried through bounded simulation, surfaced via governed interaction (XR only when it improves comprehension or rehearsal), and closed by monitoring outcomes back into the timeline. This position paper contributes: (i) a falsifiable MPDT definition that distinguishes decision-grade twins from dashboards and ungoverned copilots; (ii) a compact architectural contract that makes replayability, epistemic typing, and monitoring first-class constraints; and (iii) an evaluation agenda that treats calibration over time, worst-group reliability under shift, observation–simulation confusion, and controlled updates as acceptance-level evidence for deployment-grade systems.

1.4 Who this paper is for

MPDTs are inherently transdisciplinary, and we write for four overlapping communities. For clinicians and health-system leaders, the goal is a practical “health-centre” paradigm: a shared space where early signals, treatment options, and patient explanations can be coordinated with traceability [23, 44]. For medical informatics and data standards communities, MPDT feasibility depends on composable representations, provenance, and integration across institutional boundaries [5, 12, 35, 49, 57]. For AI/ML researchers, the challenge is not only predictive performance but calibrated uncertainty, robustness, bias-aware evaluation, and safety-aligned lifecycle monitoring in human-in-the-loop settings [1, 11, 32, 36]. For XR/HCI researchers, the central problem is interactional reliability: how embodied interfaces can reduce ambiguity, surface uncertainty, and support shared decision-making without amplifying overtrust [19, 40, 45]. Together, these perspectives motivate MPDTs as a concrete target for the future of personalized health in the metaverse: simulated, explainable care that is governed like clinical infrastructure, not consumed like media [7, 31, 52].

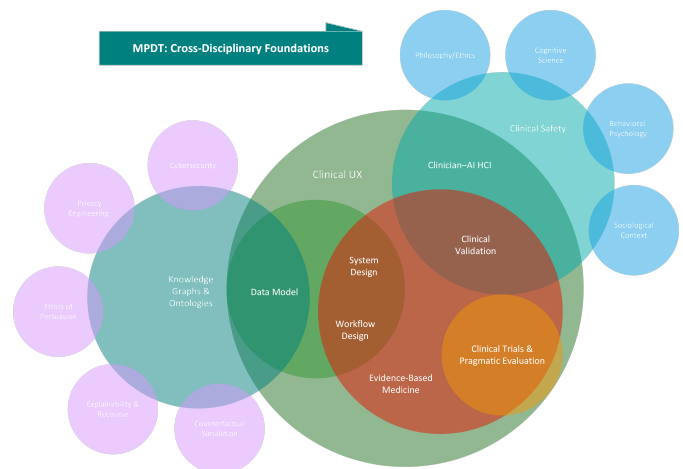


Figure 2: MPDT as a socio-technical framework context.

Figure 2 summarizes the MPDT boundary as a single socio-technical system: clinical evidence and workflows define accountable decision points; the computational twin maintains versioned state and bounded counterfactuals under uncertainty; and XR/HCI governs how stakeholders interrogate and communicate those artifacts without inducing overtrust or category errors. The enabling ring highlights deployment constraints that must be first-class: interoperability and semantics for replay across sites, governance and regulation for logging and controlled change, cybersecurity and privacy for what can be rendered and retained, ethics and consent for role-scoped purposes, and data stewardship for provenance that survives time and maintenance.

2 Why a Metaverse Patient Digital Twin

The central claim is that *patient modeling is no longer the only bottleneck*; the constraining factor is the *interaction loop* that determines how clinicians and patients interrogate state, explore alternatives, and remain responsible under uncertainty. Digital-twin reviews highlight fragmentation and uneven maturity, suggesting that having a model is insufficient unless it becomes a decision-grade artifact in workflow [21]. In parallel, XR in medicine advances rapidly but is often deployed as a communication or training overlay detached from executable, updateable patient state [19, 40]. MPDT targets the missing integration: a governed substrate where evidence, state, simulation, explanation and decisions are replayable.

2.1 Interaction is the bottleneck

Clinical decision support often fails not because a model is absent, but because its use is poorly structured. A system becomes unsafe when its limitations are easy to miss, when provenance is hard to inspect, or when what the clinician saw and acted on cannot later be reconstructed [23]. This problem becomes sharper with language-based copilots, whose outputs may sound coherent while varying in reliability across prompts and tasks [11]. The point is not simply that interfaces matter; it is that the interface partly determines the real decision process. MPDT takes that seriously. It treats interaction not as a final presentation layer, but as part of the clinical mechanism through which evidence is interpreted, simulation is used, and decisions are committed.

2.2 From prediction to counterfactual rehearsal

Most clinical decisions are counterfactual: what changes under option A versus option B. This requires twins to support *bounded* “what-if” queries anchored to a specific evidence snapshot and state version, with explicit assumptions, validity scope, and uncertainty [44]. Without a governed substrate, counterfactual capability remains locked in code or becomes persuasive storytelling. MPDT makes counterfactual rehearsal a workflow primitive: when a scenario is triggered, the system must expose inputs and assumptions, render uncertainty in a calibrated way, and emit a replayable trace that can be reviewed later [23]. Methodologically, MPDT distinguishes causal counterfactuals for interventions [39] from counterfactual explanations that support human understanding and recourse [56], and treats both as artifacts that must remain criticizable under audit.

Table 1: Why MPDT is a new systems primitive: existing paradigms fail to deliver replayable and auditable decision artifacts under deployment constraints.

Paradigm / Layer	What it already enables (<i>today</i>) vs. What remains missing (<i>MPDT target</i>)
Patient digital twins	Enables: Stateful patient-specific modeling for bounded clinical questions; increasing emphasis on lifecycle-managed, fit-for-purpose twins [4, 31, 44, 55]. Missing (MPDT): A governed interaction substrate where state, uncertainty, provenance, and updates are inspectable and replayable at decision time [14, 23, 32, 52].
XR / medical metaverse	Enables: Embodied communication and rehearsal; training/planning gains under outcome-aligned evaluation [19, 27, 40, 45, 46]. Missing (MPDT): Executable, uncertainty-first patient state under the experience; explicit observation vs. simulation separation to mitigate overtrust [23, 24, 38, 42].
Foundation-model copilots	Enables: High-coverage summarization/QA and documentation assistance; clinician-anchored evaluation patterns [2, 30, 48, 50]. Missing (MPDT): Hard grounding + auditability + change control: claims tied to evidence/state/trace IDs; monitored failure modes and bias under workflow conditions [11, 32, 34, 36].
Interoperability + semantics	Enables: Reusable schemas for exchange, imaging, terminology, and observational analytics [5, 12, 35, 49, 57]. Missing (MPDT): End-to-end provenance across derived artifacts (state, simulation, explanation); cross-site reproducibility and “replay by construction” [9, 28, 31, 57].
Clinical reporting + evidence quality	Enables: Transparent reporting for AI interventions and prediction models; risk-of-bias appraisal [3, 26, 29, 43, 58]. Missing (MPDT): MPDT-specific reporting: how interaction, simulation, and XR change decisions; logged evidence paths and confusion/error audits (obs vs. sim) [6, 23, 24, 38].
Regulation + lifecycle discipline	Enables: Risk/QMS/software lifecycle expectations and controlled evolution [15–17, 32, 54]. Missing (MPDT): Operational coupling: version registry + calibration packs + rollback; predetermined change control applied to models & interfaces, not only code [10, 32, 37, 54].
Cybersecurity + privacy	Enables: Security as a safety property; privacy-preserving analytics and multi-site learning [13, 18, 20, 41, 53]. Missing (MPDT): Threat-modeled MPDT: secure-by-design data/trace storage, role-based XR rendering, privacy leakage controls for embodied representations [13, 20, 53].
Causal + counterfactual reasoning	Enables: Formal causal counterfactuals and actionable counterfactual explanations [39, 56]. Missing (MPDT): Decision-grade counterfactual rehearsal: bounded scenarios with validity scope + uncertainty + trace IDs; comparison logged as a clinical artifact [14, 23, 52].
Monitoring + drift operations	Enables: Ongoing monitoring of calibration and dataset shift in deployment [10, 32, 37]. Missing (MPDT): MPDT monitoring that spans interaction (overtrust), simulation misuse, subgroup drift, and replay coverage—beyond model metrics alone [32, 36, 38].

2.3 Embodied explanation and training with controlled fidelity

MPDTs unify decision support and medical training under one accountability contract. Scenario-based education emphasizes repetition and situated interaction [45], while XR planning and communication motivate embodied rehearsal [19]; MPDT binds these threads by grounding virtual patient experiences in versioned state and replayable traces when personalization is required. The key commitment is *controlled fidelity*: not maximal realism, but credible, auditable utility—since richer visuals do not automatically improve decisions without validity, uncertainty, and operational constraints [31, 42]. Embodiment is therefore used only when it measurably improves comprehension, coordination, or training transfer relative to standard interfaces. Table 1 contrasts what each paradigm enables versus what remains missing, and Table 2 operationalizes the integration by specifying the planes, the artifacts each must *produce*, and the audit evidence each must *log* to support replayability, observation–simulation typing, and governed deployment.

Table 2: MPDT stack with color-coded planes; each row lists produced artifacts and required audit logs.

Plane / Layer	Produces vs. Must log
Evidence + semantics	Produces: - FHIR/OMOP-aligned longitudinal timeline - DICOM-linked imaging pointers - SNOMED-coded clinical concepts - Typed feature stores with lineage [5, 12, 35, 49] Must log: - Source provenance and transformation trail - Parameterization and context-of-collection - Dataset documentation / datasheets [9, 57]
Consent + purpose	Produces: - Use-purpose envelope (care / training / research) - Role scopes and view constraints - De-identification constraints Must log: - Consent state snapshots - Access decisions and justification metadata - Export events (who / what / when / why)
Twin state + versioning	Produces: - Versioned patient state with uncertainty fields - State deltas since last visit [31, 55] Must log: - State IDs and evidence snapshot IDs - Preprocessing / mapping hashes - Reconstruction tuple for replay [14, 23]
Forecasting + calibration	Produces: - Risk / trajectory outputs - Calibration artifacts and subgroup reports [10, 36, 37] Must log: - Per-version calibration pack (e.g., reliability/ECE) - Drift triggers and alert thresholds - Worst-group gaps [33, 34]
Simulation	Produces: - Bounded scenario runner - Treatment rehearsals traces - Sensitivity summaries [39, 52, 56] Must log: - Scenario registry and assumption sets - Parameters and validity scope - Uncertainty bands and trace IDs [14, 23]
Interaction (3D/XR)	Produces: - Linked views over Observed vs Simulated - Comparative branches and uncertainty objects [40, 45] Must log: - What was shown / clicked / compared - Category-error probes and time-to-decision - Override / deferral logs [6, 24, 38]
Copilot (governed)	Produces: - RAG summaries grounded in typed artifact IDs - Clinician/patient tailored explanations [8, 25] Must log: - Retrieved sources and grounding rate - Refusal / uncertainty behavior - Prompt + context snapshot for audit [11, 32]
Lifecycle + quality	Produces: - Change-control plan and validation reports - Rollback paths and documentation bundles [15, 16, 54] Must log: - Version registry (code + model + UI) - Test evidence and incident reports - CAPA-style logs
Security	Produces: - Secure storage and key management - Segmentation and authenticated APIs [13, 18, 53] Must log: - Threat-model updates - Access anomalies - Integrity checks for evidence/state/trace artifacts

Blue = data/consent foundations; Purple = state + prediction; Yellow = simulation; Green = interaction + copilot; Pink = lifecycle + security.

3 System Architecture

This section instantiates MPDTs as a concrete, implementable system architecture. The guiding stance is that “architecture” is not just a software diagram: it is the mechanism by which MPDTs make *clinical accountability possible*. Each layer therefore has (i) explicit responsibilities, (ii) explicit artifacts (what must be produced), and (iii) explicit validation hooks (what must be testable).

Table 3: System architecture: minimal outputs per layer and the primary validation hook each layer must support.

Layer	Minimal outputs (artifacts)	Primary validation hook
Data layer	Standards-grounded timeline; feature stores; transformation log [5, 12, 35, 49].	Provenance + data-quality checks; mapping correctness.
Twin modeling	Versioned twin state; uncertainty fields; model/version registry [31].	Analytical validation + calibration; reproducibility across versions.
Simulation	Replayable simulation traces; scenario registry [14].	Credibility testing vs. endpoints; sensitivity/assumption audits.
Interaction (XR/3D)	Linked views; role-based access; interaction logs [23].	Human-factors evaluation (error modes, overtrust, comprehension).
Copilots	Retrieval-grounded summaries/explanations; provenance links [32].	Factuality/harm checks; grounding rate; audit completeness.

Table 3 specifies MPDT as an implementable contract rather than an aspirational stack. The data layer produces standards-grounded timelines and lineage. The twin layer maintains versioned state with explicit uncertainty and exposes clinically meaningful queries. The simulation layer emits bounded scenario traces with assumptions, validity scope, and replay handles. The interaction layer renders linked views that enforce OBSERVED versus SIMULATED typing and captures exposure, overrides, and decisions. Copilots operate as governed functions over typed identifiers (evidence/state/trace IDs), making grounding and auditability measurable rather than implied [8, 25].

4 Core Use Cases

MPDTs are best judged by whether they stabilize workflows that fail today due to missing replayability, untyped simulation, and ungoverned interaction under time pressure.

4.1 Early disease detection and risk stratification in longitudinal care

The MPDT ingests new observations into the standards-grounded timeline, updates the versioned twin state, and re-computes risk trajectories with uncertainty. Clinicians interrogate *why* risk changed via linked evidence and deltas. If warranted, they run bounded counterfactuals (e.g., “if adherence improves” or “if a confound is removed”) and compare projected branches before deciding on testing, monitoring, or referral. The system logs what evidence and scenarios were used and what action was taken, enabling later audit and outcome association [12, 23, 35]. In MPDT terms, the scientific object is not the risk score but the replayable trajectory of evidence → state → counterfactuals → action, enabling auditable attribution when triage outcomes drift or disagree across sites.

4.2 Diagnosis support and explanation: clinician-facing and patient-facing

The clinician-facing pathway begins with the twin state and uncertainty, then drills down into evidence contributions and differential hypotheses under explicit validity scope. The patient-facing pathway uses the same substrate but renders simplified linked views: observed facts, model-supported inferences, and simulated projections are separated and labeled, with patient questions answered by a copilot constrained to evidence/state/trace IDs [23, 32]. The validation target is interpretation correctness under time pressure: distinguishing observed facts from simulated outcomes, and reducing automation-induced error modes [24, 38]. The MPDT contribution is that explanation becomes a typed, replayable view over evidence/state/trace IDs (not a narrative), so disagreement and harm can be investigated as a reconstruction problem rather than a subjective interpretation problem.

4.3 Treatment planning and counterfactual rehearsal

Treatment planning becomes a controlled comparison of simulation traces. The clinician, GP, or specialist team triggers bounded simulations from the same anchored twin state, inspects comparative branches with uncertainty, and records decisions with rationale linked to evidence and trace IDs. MPDTs insist that simulated outcomes are never presented without their trace and validity scope, to prevent persuasive error where plausible simulations are treated as facts [23, 52]. The research object is the logged comparison of scenario traces with explicit validity envelopes and uncertainty, turning “what-if” reasoning into a contestable clinical artifact rather than descriptive simulation output.

4.4 Medical training with controllable virtual patients grounded in twin state

In training mode, the MPDT instantiates virtual patient scenarios that are either synthetic but constrained by plausible state dynamics, or personalized to a de-identified twin state when permitted. Trainees interact through structured tasks while the system logs actions, evidence accessed, and branches chosen; educators replay sessions and assess competency [22, 45]. Transfer beyond the simulator is the evaluation bar [27, 46]. MPDT makes training and care compatible by reusing the same artifact contracts (state versions, scenario traces, interaction logs), so transfer and safety can be evaluated with the same replay and typing guarantees rather than separate bespoke simulators.

4.5 Cross-site coordination: MPDT as a shared hub across departments and sites

The MPDT acts as a shared coordination object: a versioned twin state provides a consistent reference for multidisciplinary teams, while role-based interfaces provide tailored views. Handoffs become auditable: what was known at transfer time, what simulations were considered, and what uncertainty remained unresolved. Monitoring services aggregate drift signals and workflow outcomes across sites, enabling controlled updates and rollback under governance rather than ad-hoc changes [12, 17, 32, 35]. In this setting, MPDT

provides a common, reconstructible basis for handoff, comparison, and controlled change across sites.

5 Governance and Evaluation: What Must Be Measured to Deploy

Governance is therefore an evaluation problem: the system must continuously produce evidence of validity for its stated scope, of calibration in deployment, and of whether interaction induces systematic errors such as treating simulations as observations [23, 32]. A key stance is that logs are not merely by-products; they are the evaluation substrate. When evidence snapshots, state versions, scenario traces, and interaction events are typed and replayable, the system can compute acceptance-level signatures: replay coverage for decisions, observation–simulation confusion rate under time pressure, rolling calibration stability, worst-group reliability under shift, and controlled update behavior linked to explicit version registries.

Success for this idea is achievable. In a successful MPDT deployment, post-hoc reconstruction becomes routine, simulations remain epistemically safe because they are typed, bounded, and open to critique, and trust is measurable through calibration and worst-group reliability over time. If MPDT succeeds, the community gains a shared systems primitive for accountable, counterfactual, human-facing decision support, enabling rigorous research on interaction policies, simulation validity envelopes, and lifecycle governance as first-class scientific objects rather than afterthoughts.

6 Conclusion

This paper argues that the next leap in personalized, explainable care will come from closing the loop between patient modeling and patient interaction. Digital twins, XR medicine, and clinical AI copilots each demonstrate value, but typical deployments remain siloed: models are not replayable across time, XR risks descriptive visualization detached from executable state, and copilots can be fluent without auditable grounding. MPDT reframes the problem as a systems primitive: a versioned patient state coupled to bounded counterfactual simulation under explicit uncertainty, exposed through governed interaction that enforces OBSERVED versus SIMULATED typing, and paired with audit artifacts that make monitoring and controlled change feasible.

The immediate research opportunity is to treat interaction policy, simulation validity envelopes, and lifecycle governance as algorithmic objects with measurable acceptance criteria. A limitation of this position is that it specifies the contract and acceptance evidence rather than a full instantiation; the Blue-Sky challenge is to develop methods and benchmarks that learn, optimize, and audit these artifact-graphs at scale (across sites, subgroups, and shifting data regimes). If successful, MPDT shifts KDD’s frontier toward systems where the *unit of discovery* is a replayable decision artifact—enabling rigorous science on what interaction and simulation policies reduce harm, improve workflow outcomes, and remain reproducible under controlled evolution.

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